The documentation and process conversion measures necessary to comply with this document shall be completed by 10 June 2005.

INCH-POUND

MIL-PRF-19500/581B
10 March 2005
SUPERSEDING
MIL-PRF-19500/581A
10 September 1999

* PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON AMPLIFIER, TYPES 2N4237, 2N4238, AND 2N4239, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, amplifier transistors. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, (TO-39).

* 1.3 Maximum ratings, unless otherwise specified $T_A$ = +25°C.

| Type | $P_T$ $T_A$ = +25°C (1) | $P_T$ $T_C$ = +25°C (2) | $R_{\theta JA}$ | $R_{\theta JC}$ | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ | $I_B$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | °C/W | V dc | V dc | V dc | A dc | A dc | °C |
| 2N4237 | 1.0 | 6.0 | 175 | 18 | 50 | 40 | 6.0 | 1.0 | 0.5 | -65 to |
| 2N4238 | 1.0 | 6.0 | 175 | 18 | 80 | 60 | 6.0 | 1.0 | 0.5 | +200 |
| 2N4239 | 1.0 | 6.0 | 175 | 18 | 100 | 80 | 6.0 | 1.0 | 0.5 | |

(1) See figure 2.
(2) See figure 3.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

AMSC N/A FSC 5961

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1.4 Primary electrical characteristics, unless otherwise specifed at $T_A$ = +25°C.

| Limits | $h_{FE}$ at $V_{CE}$ = 1.0 V dc (1) | | | $\lvert h_{fe} \rvert$ f = 10 MHz | $C_{obo}$ f = 1.0 MHz | $V_{CE(sat)2}$ 1/ | $V_{BE(sat)2}$ 1/ |
|---|---|---|---|---|---|---|---|
| | $h_{FE1}$ $I_C$ = 100 mA dc | $h_{FE2}$ $I_C$ = 250 mA dc | $h_{FE3}$ $I_C$ = 500 mA dc | $V_{CE}$ = 10 V dc $I_C$ = 100 mA dc | $V_{CB}$ = 10 V dc $I_C$ = 0 | $I_C$ = 1.0 A dc $I_B$ = 0.1 A dc | $I_C$ = 1.0 A dc $I_B$ = 0.1 A dc |
| Min | 30 | 30 | 30 | 3.0 | pF | | V dc |
| Max | | 150 | | | 100 | 0.6 | 1.5 |

(1) Pulsed see 4.5.1.

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

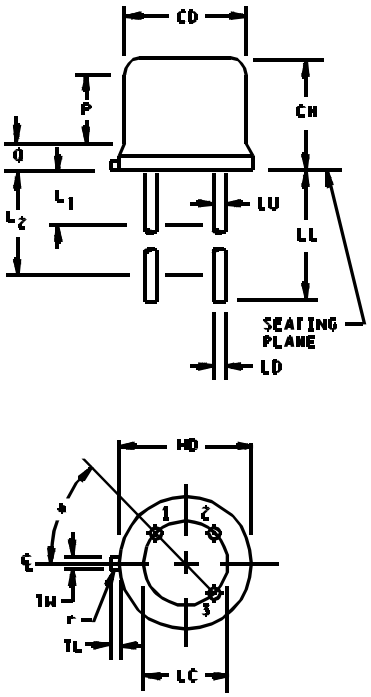
3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).



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| Ltr | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .305 | .355 | 7.75 | 9.02 | |
| CH | .240 | .260 | 6.10 | 6.60 | |
| HD | .335 | .370 | 8.51 | 9.40 | |
| LC | .200 TP | | 5.08TP | | 6 |
| LD | .016 | .021 | 0.41 | 0.53 | 7 |
| LL | .500 | .750 | 12.70 | 19.05 | 7 |
| LU | .016 | .019 | 0.41 | 0.48 | 7 |
| L1 | | .050 | | 1.27 | 7 |
| L2 | .250 | | 6.35 | | 7 |
| TL | .029 | .045 | 0.74 | 1.14 | 3 |
| TW | .028 | .034 | 0.71 | 0.86 | 10 |
| P | .100 | | 2.54 | | 5 |
| Q | | .050 | | 1.27 | 4 |
| R | | .010 | | 0.25 | 11 |
| α | 45° TP | | 45° TP | | 6 |
| Notes | 1, 2, 8, 9 | | | | |

NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Symbol CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) relative to tab. Device may be measured by direct methods or by gauge.
7. Symbol LD applies between $L_1$ and $L_2$. Dimension LD applies between $L_2$ and LL minimum.
8. Lead designation, depending on device type, shall be as follows: 1 Emitter, 2 Base, and 3 Collector.
9. Lead number three is electrically connected to case.
10. Beyond r maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
11. Symbol r applied to both inside corners of tab.
12. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 1. Physical dimensions (TO-39).



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* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

| | |
|---|---|
| $R_{\theta JA}$ | Thermal resistance junction to ambient. |
| $R_{\theta JC}$ | Thermal resistance junction to case. |
| TRB | Technical review board. |

3.4 Interface and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (TO-39) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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* 4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table IV of MIL-PRF-19500) | Measurement |
|---|---|
| | JANTX and JANTXV level |
| (1) 3c | Method 3131 (see 4.5.3) |
| 7 | Optional |
| 11 | $I_{CBO}$, and $h_{FE2}$ |
| 12 | See 4.3.1. |
| 13 | Subgroup 2 of table I herein<br>$\Delta I_{CBO}$ = 100 percent of initial value, or 10 nA dc whichever is greater;<br>$\Delta h_{FE2}$ = ± 15 percent of initial value. |
| 14 | Required |

(1) Shall be performed anytime before screen 10.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $V_{CB}$ = 10 - 30 V dc. Power shall be applied to achieve $T_J$ = +135°C minimum using a minimum $P_D$ = 75 percent of $P_T$ maximum, $T_A$ ambient rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, TJ, and mounting conditions) may be used. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval. This option is limited to plants who are at least transitional (QML) approved or have an approved technical review board (TRB).

* 4.3.2 Thermal impedance measurements. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$ (and $V_C$ where appropriate). The thermal impedance limit used in 4.3, screen 3c and subgroup 2 of table I shall comply with the thermal impedance graph in figures 4 and 5 (less than or equal to the curve value at the same $t_H$ time) and shall be less than the process determined statistical maximum limit as outlined in method 3131 of MIL-STD-750.

* 4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of subgroups 1 and 2 of table I herein, inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2 herein).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. End-point electrical measurements shall be in accordance with subgroup 2 of table I herein.



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* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500, and herein. Electrical measurements (end-points) and delta requirements shall be after each step in 4.4.2.1 and shall be in accordance with table I, subgroup 2 and 4.5.3 herein.

* 4.4.2.1 Group B inspection, appendix E, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500. Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action.

| Step | Method | Condition |
|---|---|---|
| 1 | 1027 | Steady-state life: 1,000 hours minimum, $V_{CB}$ = 10 V dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = +150°C, $V_{CB}$ = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +200°C. n = 22, c = 0. |

4.4.2.2 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV, samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroup B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with 4.5.3 herein.

4.4.3.1 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E. |
| C5 | 3131 | See 1.3. |
| C6 | 1026 | $V_{CB} \geq$ 10 V dc; adjust $P_T$ to achieve $T_J$ = + 175 °C minimum; $T_A$ = + 30°C ± 5°C. No heat sink or forced-air cooling on the device shall be permitted. |

* 4.4.3.2 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.



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* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with the applicable steps of 4.5.3 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

* 4.5.3 Delta requirements. Delta requirements shall be as specified below:

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Collector to base cutoff current | 3036 | Bias condition D | $\Delta I_{CBO}$ (1) | 100 percent of initial value or 10 nA dc whichever is greater. | | |
| | 2N4237<br>2N4238<br>2N4239 | | $V_{CB}$ = 50 V dc<br>$V_{CB}$ = 80 V dc<br>$V_{CB}$ = 100 V dc | | | | |
| 2. | Forward current transfer ratio | 3076 | $I_C$ = 250 mA dc<br>$V_{CE}$ = 1.0 V dc;<br>Pulsed (see 4.5.1) | $\Delta h_{FE2}$ (1) | ± 25 percent change from initial recorded value. | | |

(1) Devices which exceed the table I limits for this test shall not be shipped.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical 3/ examination | 2071 | n = 45 devices, c = 0 | | | | |
| Solderability 3/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to 3/ 4/ solvent | 1022 | n = 15 devices, c = 0 | | | | |
| Temp cycling 3/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Heremetic seal | 1071 | n = 22 devices, c = 0 | | | | |
| Fine leak<br>Gross leak | | | | | | |
| Electrical measurements | | Table I, subgroup 2 | | | | |
| Bond strength 3/ | 2037 | Precondition $T_A$ = +250°C at t = 24 hrs or $T_A$ = +300°C at t = 2 hrs, n = 11 wires, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3131 | See 4.3.2 | $Z_{\theta JX}$ | | | °C/W |
| Breakdown voltage collector to emitter<br>2N4237<br>2N4238<br>2N4239 | 3011 | Bias condition D, pulsed (see 4.5.1) $I_C$ = 100 mA dc | $V_{(BR)CEO}$ | <br>50<br>80<br>100 | | Vdc |
| Collector emitter cutoff current<br>2N4237<br>2N4238<br>2N4239 | 3041 | Bias condition A, $V_{BE}$ = 1.5 V dc<br>$V_{CE}$ = 50 V dc<br>$V_{CE}$ = 80 V dc<br>$V_{CE}$ = 100 V dc | $I_{CEX1}$ | | 100 | nA dc |
| Collector to baser cutoff current<br>2N4237<br>2N4238<br>2N4239 | 3036 | Bias condition D<br>$V_{CE}$ = 50 V dc<br>$V_{CE}$ = 80 V dc<br>$V_{CE}$ = 100 V dc | $I_{CBO}$ | | 100 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{BE}$ = 6 V dc | $I_{EBO}$ | | 0.5 | mA dc |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Forward current transfer ratio | 3076 | Pulsed (see 4.5.1), $I_C$ = 100 mA dc<br>$V_{CE}$ = 1.0 V dc | $h_{FE1}$ | 30 | | |
| Forward current transfer ratio | 3076 | Pulsed (see 4.5.1), $I_C$ = 250 mA dc<br>$V_{CE}$ = 1.0 V dc | $h_{FE2}$ | 30 | 150 | |
| Forward current transfer ratio | 3076 | Pulsed (see 4.5.1), $I_C$ = 500 mA dc<br>$V_{CE}$ = 1.0 V dc | $h_{FE3}$ | 30 | | |
| Collector to emitter voltage (saturated) | 3071 | Pulsed (see 4.5.1), $I_C$ = 500 mA dc<br>$I_B$ = 50 mA dc | $V_{CE(sat)1}$ | | 0.3 | V dc |
| Collector to emitter voltage (saturated) | 3071 | Pulsed (see 4.5.1), $I_C$ = 1.0 A dc<br>$I_B$ = 0.1 A dc | $V_{CE(sat)2}$ | | 0.6 | V dc |
| Base emitter voltage | 3066 | Test condition A, pulsed (see 4.5.1), $I_C$ = 500 mA dc, $I_B$ = 50 mA dc | $V_{BE(sat)1}$ | | 1.0 | V dc |
| Base emitter voltage | 3066 | Test condition A, pulsed (see 4.5.1), $I_C$ = 1.0 A dc, $I_B$ = 0.1 A dc | $V_{BE(sat)2}$ | | 1.5 | V dc |
| Subgroup 3 | | | | | | |
| High-temperature operation: | | $T_A$ = +150°C | | | | |
| Collector to emitter cutoff current<br>2N4237<br>2N4238<br>2N4239 | 3041 | Bias condition A, $V_{BE}$ = 1.5 V dc<br>$V_{CE}$ = 30 V dc<br>$V_{CE}$ = 50 V dc<br>$V_{CE}$ = 70 V dc | $I_{CEX2}$ | | 25 | µA dc |
| Low-temperature operation: | | $T_A$ = -55°C | | | | |
| Forward current transfer ratio | 3076 | Pulsed (see 4.5.1), $I_C$ = 250 mA dc<br>$V_{CE}$ = 1.0 V dc | $h_{FE4}$ | 15 | | |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 | | | | | | |
| Magnitude of small-signal short-circuit forward-current transfer ratio | 3306 | $I_C$ = 100 mA dc, $V_{CE}$ = 10 V dc, f = 10 MHz | $\|h_{FE}\|$ | 3 | | |
| Open circuit output capacitance | 3236 | $I_E$ = 0, $V_{CB}$ = 10 V dc, f = 100 kHz | $C_{obo}$ | | 100 | pF |
| Subgroup 5 | | | | | | |
| Safe operating area (continuous dc) 5/ | 3051 | $T_C$ = +25°C; t ≥ 0.5 s, 1 cycle. | | | | |
| Test 1 | | $I_C$ = 1.0 A dc, $V_{CE}$ = 6 V dc | | | | |
| Test 2 | | $I_C$ = 500 mA dc, $V_{CE}$ = 12 V dc | | | | |
| Test 3 | | | | | | |
| 2N4237 | | $I_C$ = 166 mA dc, $V_{CE}$ = 30 V dc | | | | |
| 2N4238 | | $I_C$ = 100 mA dc, $V_{CE}$ = 50 V dc | | | | |
| 2N4239 | | $I_C$ = 71 mA dc, $V_{CE}$ = 70 V dc | | | | |
| End-point electrical measurements | | See table I, subgroup 2 herein. | | | | |
| Subgroups 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1, double the sample size of the failed test or sequence of tests.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ L = 5 mH (2 each Essex Stancor C-2688 in parallel 1A, 0.5 ohm, or equivalent (see 4.5.2).



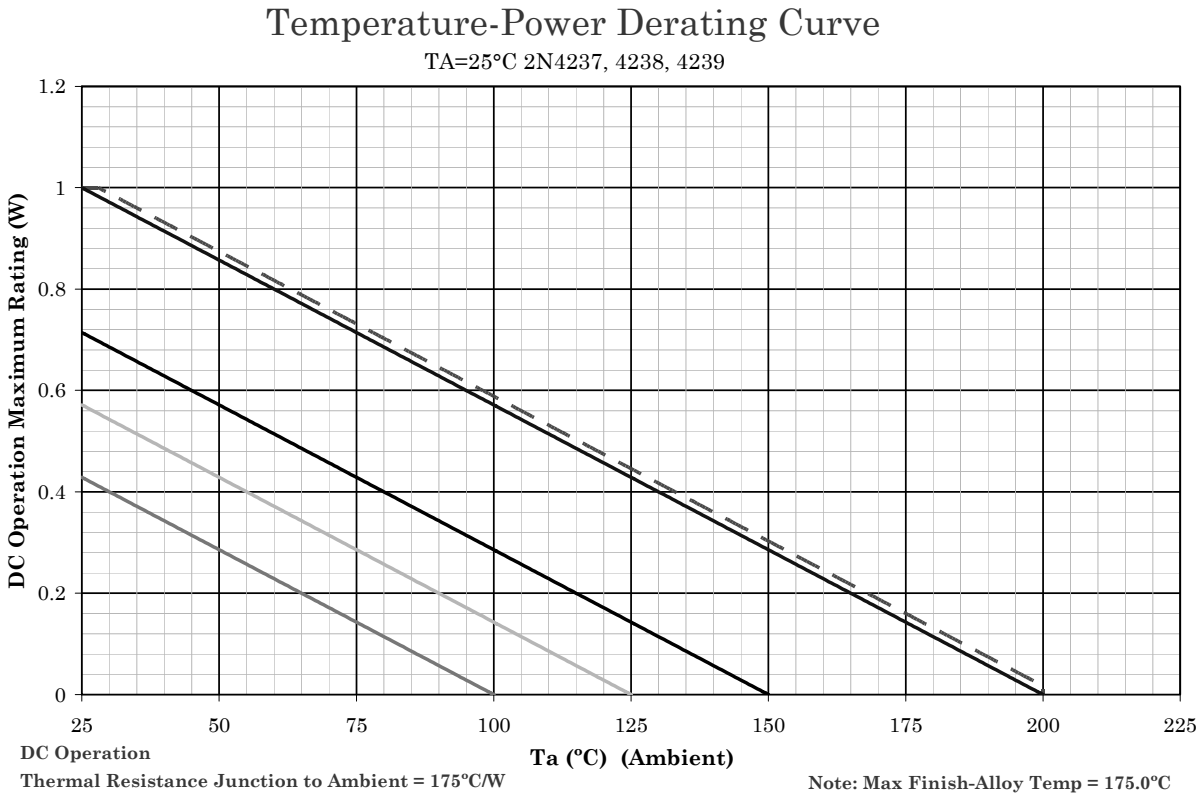
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* TABLE II. Group E inspection (all quality levels) - for qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Intermittent life | 1037 | $V_{CB}$ = 10 V dc, 6,000 cycles, forced air cooling allowed on cooling cycle only. | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 4 | | | |
| Thermal impedance curves | | Each supplier shall submit their qualification lot average and design maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report. | Sample size<br>N/A |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition B. | |



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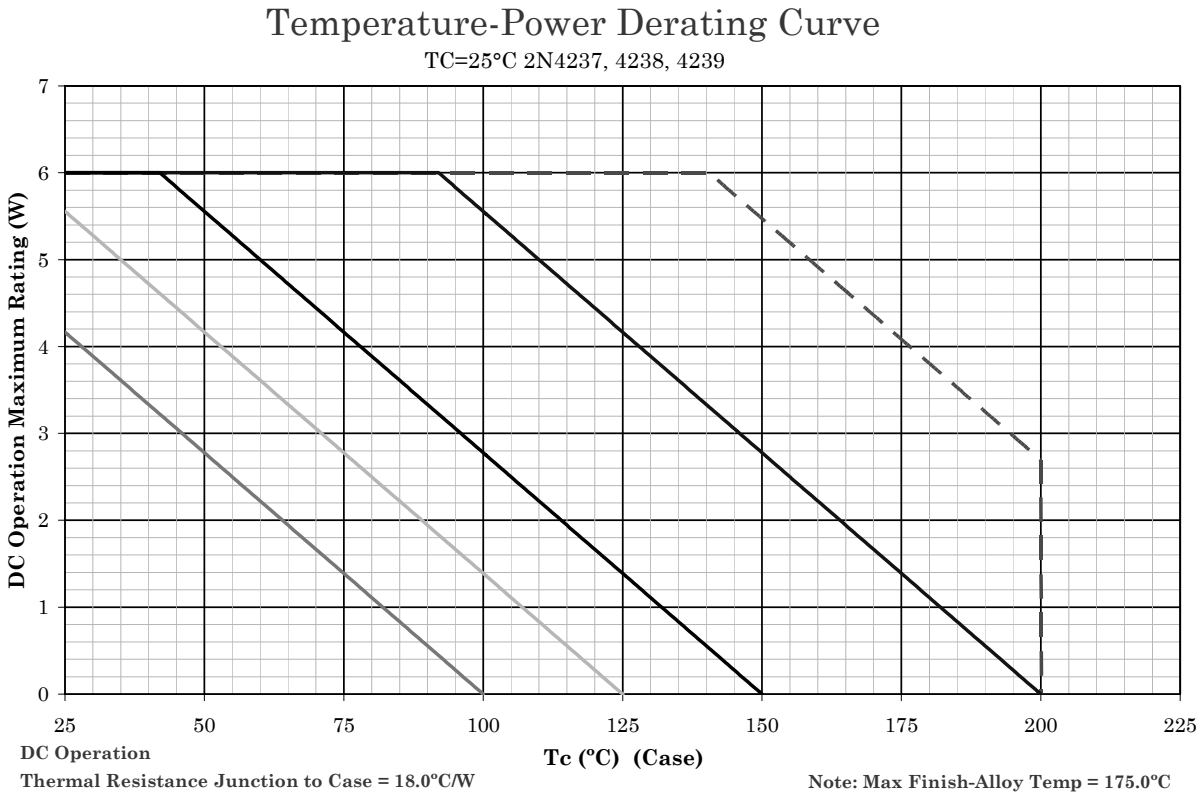

NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 2. Derating for 2N2060 ($R_{\theta JA}$) (TO-39).



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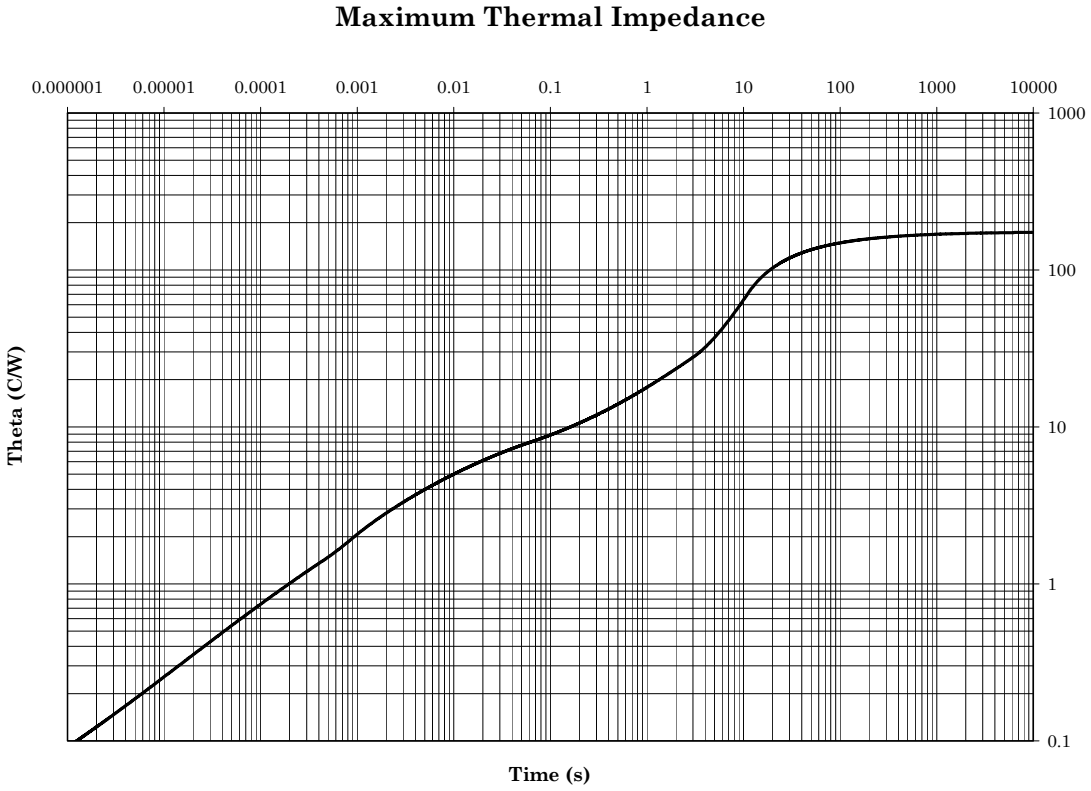

NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 3. Derating for 2N2060 ($R_{\theta JC}$) (TO-39).



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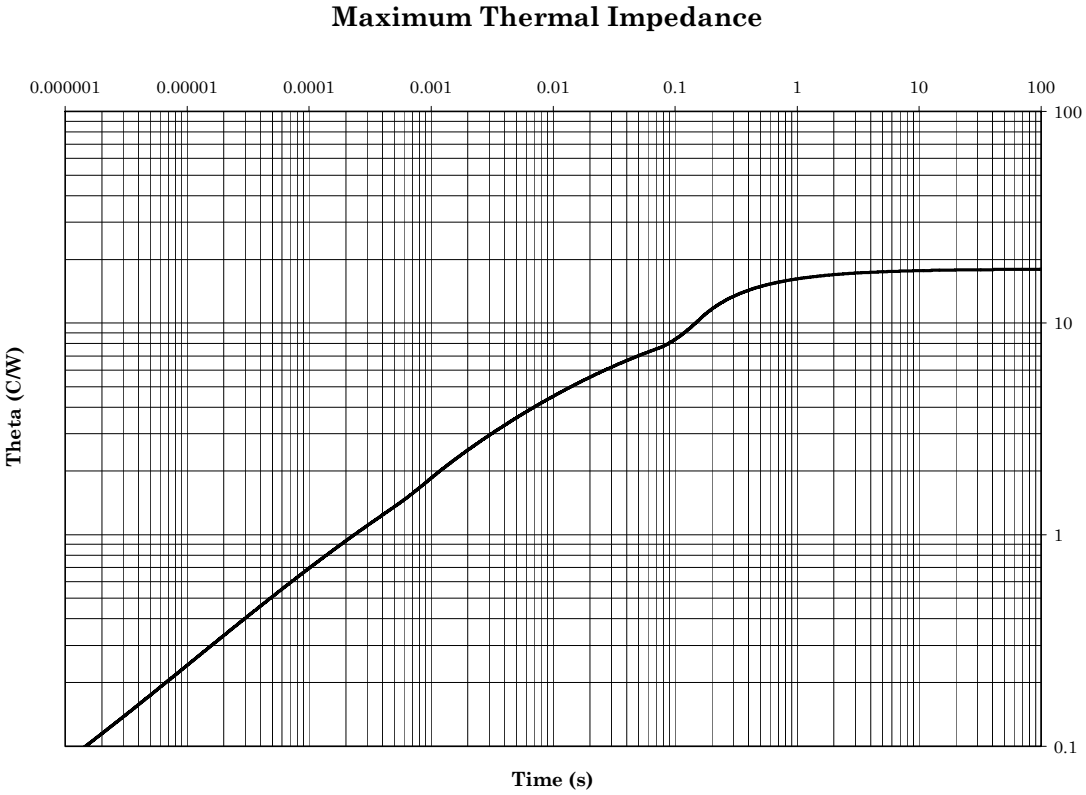



$T_A$ = + 25°C, thermal resistance = 175°C/W.

* FIGURE 4. Thermal impedance graph ($R_{\theta JA}$) for 2N2060 (TO-39).



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$T_C$ = + 25°C, thermal resistance = 18°C/W.

* FIGURE 5. Thermal impedance graph ($R_{\theta JC}$) for 2N2060 (TO-39).



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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Application guidance. The following NPN type transistor is complementary to the PNP device listed herein.

| NPN | PNP |
|---|---|
| 2N4237 | 2N4234 |
| 2N4238 | 2N4235 |
| 2N4239 | 2N4236 |



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6.5 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable for the military PIN.

| Military PIN | Manufacturer's CAGE Code | Manufacturer's and user's PIN |
|---|---|---|
| JAN2N4237 or JANTX2N4237 or JANTXV2N4237 | 43611 | 2N4237<br>ST1027H |
| JAN2N4238 or JANTX2N4238 or JANTXV2N4238 | 43611 | 2N4238<br>ST1375H |
| JAN2N4239 or JANTX2N4239 or JANTXV2N4239 | 43611 | 2N4239<br>ST690H2<br>ST693H1<br>ST693H2<br>ST693H3<br>ST693H4<br>ST693H32<br>ST7008H1<br>ST7008H2<br>ST7008H3<br>ST7008H4<br>ST7008H21<br>ST7008H32 |

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project 5961-2917)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/ .



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